

Physics 7501 Quiz 1
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(1.1.3) Pauli matrices

Recall the Pauli matrices

$$\hat{X} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \hat{Y} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \hat{Z} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (1.1)$$

Express the following matrix $\hat{M}$ as a linear superposition of identity matrix $\hat{1}_{2 \times 2}$ and Pauli matrices:

$$\hat{M} = \begin{pmatrix} 3 & 6 \\ 4 & 5 \end{pmatrix} = m_0 \hat{1}_{2 \times 2} + m_1 \hat{X} + m_2 \hat{Y} + m_3 \hat{Z}. \quad (1.2)$$

Obtain $m_{0,1,2,3}$.

For $\hat{\sigma}_0 = \hat{1}, \hat{\sigma}_1 = \hat{X}, \hat{\sigma}_2 = \hat{Y}, \hat{\sigma}_3 = \hat{Z}$, note that under the Hilbert-Schmidt norm $\langle \psi_i | \psi_j \rangle = \text{tr}(\psi_i^\dagger \psi_j) = 2\delta_{ij}$. Thus $m_i = \frac{1}{2} \langle \hat{\sigma}_i | \hat{M} \rangle \implies$

$$m_0 = \frac{1}{2} \text{tr} \left[\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 3 & 6 \\ 4 & 5 \end{pmatrix} \right] = 4$$

$$m_1 = \frac{1}{2} \text{tr} \left[\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 3 & 6 \\ 4 & 5 \end{pmatrix} \right] = 5$$

$$m_2 = \frac{1}{2} \text{tr} \left[\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 3 & 6 \\ 4 & 5 \end{pmatrix} \right] = i$$

$$m_3 = \frac{1}{2} \text{tr} \left[\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 3 & 6 \\ 4 & 5 \end{pmatrix} \right] = -1$$